

EMILY STRICKLAND

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SUMMARY

Master's student in Data Science with internship experience at NASA's Jet Propulsion Laboratory (JPL) analyzing and validating large, complex datasets. Skilled in Python, R, and SQL, with experience in statistical analysis, machine learning, and data visualization, and in building data pipelines and warehouses with Airflow, dbt, and PostgreSQL. Interested in applying data science to impactful, data-driven work.

EDUCATION

Master of Science (MS) in Data Science | Willamette University - Salem, OR
Bachelor of Science (BS) in Data Science and Minor in Computer Science

Aug 2022 – Aug 2026

- Relevant Coursework: Advanced Data Engineering, Advanced Machine Learning, Data Visualization, Python for Data Science, Geographic Information Systems (GIS), Intro to Data Science with R

EXPERIENCE

Data Science Intern | Clinical Psychology Practice – Remote

May 2026 – July 2026

- Built Python (pdfplumber) and R workflows to extract, clean, and de-identify patient questionnaire data from PDF records into analysis-ready datasets.
- Produced visualizations comparing assessment results across patient subgroups to support the practice's research goals.

Project – Data Warehouse & Pipeline | Willamette University – Salem, OR
Advanced Data Engineering

Summer 2026

- Built a PostgreSQL data warehouse on a shared Linux VM with Docker Compose and persistent storage, implementing a star-schema dimensional model (fact and dimension tables) for a metro ridership simulation.
- Developed Apache Airflow pipelines to snapshot source data into an S3-compatible data lake as Parquet and load it into warehouse staging tables, debugging file-read and database-connection errors so every table loaded reliably.
- Wrote dbt models to transform staging data into dimension tables with reproducible, re-runnable loads, using hashed surrogate keys for stable joins.
- Built a Grafana dashboard of ridership by line and hour of day to recommend how a transit authority should allocate its limited active trains.

Data Science Summer Intern | NASA Jet Propulsion Laboratory (JPL) - Pasadena, CA

May 2025 – July 2025
May 2024 – July 2024

- Automated and improved Python scripts for aerosol dataset validation, increasing processing efficiency and accuracy.
- Conducted data verification and quality review of remote sensing products to ensure consistency and reliability across datasets.
- Developed a pyrocumulonimbus (PyroCb) match database and generated color Geostationary Operational Environmental Satellites (GOES) imagery to support Brown Carbon (BrC) analysis.
- Analyzed Black Carbon and Brown Carbon aerosol evolution in smoke plumes, integrating ancillary weather and temperature data, using R, to strengthen results.

SKILLS AND SOFTWARE

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|------------------------|------------------------------|------------------------|-----------------------|
| • Python | • Machine Learning | • ETL / Data Pipelines | • VS Code |
| • R | • Predictive Modeling | • Apache Airflow | • Generative AI Tools |
| • SQL / PostgreSQL | • Data Visualization | • dbt | • ArcGIS / Geospatial |
| • Statistical Analysis | • Data Cleaning & Validation | • Data Warehousing | • Microsoft Office |

